CodingWise

CodingWise Advantage

Land your dream development job with
AI-powered skills and guaranteed placement support

INTRODUCING

AI Powered Full Stack Development (MERN)

This isn't another theory-heavy certification. This is job focused, live mentorship, real-world projects, placement support, and guaranteed job opportunities with 800+ hiring partners.

Your Journey Ahead



Key Steps to Get Hired Faster

Learn What Others Don’t

Master in-demand AI tools , which will power your development skills.

Train Like You’re Already Hired

Mock Interviews , placement mentorship weekly.

Get Direct Job Opportunities Access

Skip job portal, Get exclusive job openings along with our 800+ hiring partners.

Build Projects That Speak for You

Create Capstone projects after each module that recruiters can't ignore - guided by industry mentors.

HTML

1. Introduction & Basics

- [x] Getting Started with HTML
- [x] What is HTML? / HTML Web Design Basics
- [x] HTML Paragraphs, Headings, Comments
- [x] HTML Tables (table, thead, tbody, tr, td, th)
- [x] HTML Lists: unordered, ordered, description list
- [x] Line break (
), Pre tag (<pre>), Horizontal rule (<hr>)
- [x] Block vs Inline elements (concept + examples)
- [x] HTML Links (<a>) — absolute vs relative paths
- [x] Images (<img>): attributes, srcset, alt, accessibility notes
- [x] Text formatting: strong, em, u, small, subscript/superscript

2. Head & Metadata

- [x] <head> structure: title, meta, charset, viewport
- [x] <meta> elements: description, keywords, og: tags (social)
- [x] Favicon setup and best practices
- [x] Inline <style> vs external CSS linking

3. Forms & Inputs

- <form> element, action & method
- Input types: text, email, password, number, date, file, hidden, checkbox, radio
- Select, textarea, label usage
- Form validation basics (HTML5 validation attributes)
- Accessibility for forms (aria, labels)

4. Semantic HTML & Accessibility

- Semantic tags: header, nav, main, section, article, aside, footer, figure, figcaption
- Landmark roles & screen reader considerations
- Best practices for accessible content

5. Graphics & Media

- HTML Video & Audio (controls, streaming considerations)
- SVG basics & inline SVG usage
- <canvas> fundamentals (draw, animations)
- Image optimization (srcset, lazy loading)

6. HTML Miscellaneous

- Iframes, Embeds
- HTML Entities, Emojis, NBSP
- File paths, linking assets

CSS

7. Introduction & Syntax

- What is CSS, how to include (inline, internal, external)
- CSS syntax, specificity, cascade principles
- CSS units (px, em, rem, %, vh/vw)

8. Selectors & Combinators

- Type selectors, class, id, attribute selectors
- Descendant, child, adjacent sibling selectors
- Pseudo-classes (:hover, :focus, :nth-child), pseudo-elements (::before, ::after)
- !important and when NOT to use it

9. Typography & Text Styling

- font-family, font-face (custom fonts)
- font-size, font-weight, font-stretch
- text-align, text-decoration, letter-spacing, line-height
- text-shadow, vertical-align

10. Box Model & Layout Fundamentals

- Box model: content, padding, border, margin
- width/height, box-sizing property
- display types (block, inline, inline-block, none)
- position (static, relative, absolute, fixed, sticky), z-index, overflow

11. CSS - Layout Systems

- Floats & clears (legacy)
- Flexbox: containers, items, alignment, responsive patterns
- CSS Grid: grid container, grid-template, placing items, responsive grids
- Use cases: galleries, dashboard layouts

12. CSS - Responsive Design & Media Queries

- Mobile-first approach, breakpoints, @media usage
- Responsive images & fluid typography
- Container queries (if available) / progressive enhancement

13. CSS - Transitions, Transforms & Animations

- transition basics, easing, delays
- transform: translate, rotate, scale, skew
- Keyframe animations, performance considerations

14. CSS - Styling Components & UI Patterns

- Buttons, forms, inputs, tables styling
- Icons and image styles, responsive navbars, card components
- Introduction to UI libraries: Tailwind CSS, Material UI, Ant Design (usage + when to pick)

15. CSS - Projects (Applied)

- Business Landing Page (responsive)
- Portfolio Website — componentized, accessible

JavaScript

16. JavaScript - Core Fundamentals

- Getting started: console.log, devtools basics
- Variables (let, const), data types, type coercion
- Operators, expressions, template literals

17. JavaScript - Control Flow & Functions

- Conditionals (if/else, ternary), switch/case
- Loops: for, while, for...of, for...in
- Functions: declarations, expressions, arrow functions, IIFEs
- Scope, hoisting, closures, recursion

18. JavaScript - Objects & Data Structures

- Plain objects, object methods, this keyword
- Prototypes, prototype inheritance, constructor functions
- Arrays (methods: map, filter, reduce), multi-dimensional arrays
- Sets, Maps, WeakMap, WeakSet

19. JavaScript - ES6+ & Modern Features

- Destructuring, spread/rest operator, default params
- Modules (import / export)
- Classes, inheritance, generators & iterators, proxies

20. JavaScript - Asynchronous Programming

- Callbacks, Promises, async/await
- fetch API, XHR basics
- Debouncing, throttling, timers (setTimeout, setInterval)

21. JavaScript - DOM Manipulation & Events

- DOM tree, selecting elements (getElementById, querySelector)
- Traversal (parent, children, siblings)
- Creating/appending/removing nodes (createElement, appendChild, innerHTML)
- Event handling: addEventListener, event object, delegation, bubbling & capturing
- Page lifecycle events (DOMContentLoaded, load)

22. Browser APIs & Storage

- Web Storage: cookies, localStorage, sessionStorage
- IndexedDB basics for offline apps
- Geolocation API, Notifications API, Push API basics
- History API (pushState) and single-page navigation patterns
- Fetch API and working with JSON

23. Misc Browser APIs & Tools

- Drag & Drop API + FileReader for uploads
- Dialog API, Popover API, Pointer events
- BOM: window, screen, navigator, location, history

24. JavaScript Libraries & Helpers

- Using Axios vs fetch, utility functions, Lodash basics
- AOS (Animation on Scroll), SwiperJS / carousel libraries

25. Authentication & Firebase (From PDF Extras)

- Firebase OTP Authentication (setup & flow)
- Auth best practices, token handling

26. Combined Frontend Project(S)

- Fully Functional Business Website (HTML/CSS/JS)
- Fully Functional News Application (client side features, search, pagination)

MERN

Module 1: MERN - Introduction & Setup

- Overview: MongoDB, Express, React, Node — why MERN
- Setting up dev environment: Node, npm/yarn, VS Code, MongoDB Atlas setup, Postman

Module 2: JavaScript For MERN & ES6+ (Reiterated, Applied To Backend & Frontend)

- Core JS, advanced features, asynchronous patterns (applied to Node & frontend)

Module 3: MongoDB

- NoSQL vs SQL concepts
- Installing MongoDB / MongoDB Atlas
- CRUD operations: insertOne/insertMany, find, update, delete
- Query operators, filtering, projection, sorting, pagination
- Aggregation framework, pipelines (group, match, project, unwind)
- Indexing (single, compound), explain plan basics
- Relationships: embedding vs referencing, data modelling patterns
- Transactions (multi-document) overview

Module 4: Node.Js (Detailed)

- Node architecture: event loop, non-blocking I/O
- Core modules: fs, path, http, os, stream basics
- npm & package management, semantic versioning, lockfiles
- Error handling and logging best practices (winston, pino)
- Process & environment variables, dotenv

Module 5: Express.Js (Detailed)

- Project structure & express app creation
- Routing: static, dynamic, param, nested routers
- Middleware: body parsing, CORS, cookie parser, custom middleware
- Request/response lifecycle, headers, status codes
- REST API design and status conventions (CRUD)
- Integration with MongoDB via Mongoose: schemas, models, validation
- Testing APIs (Postman, automated tests — supertest)

Module 6: React.Js (Detailed)

- SPA concepts, virtual DOM, JSX syntax
- Component patterns: functional components, props, state
- Hooks: useState, useEffect, useRef, useReducer, custom hooks
- Routing with React Router (nested routes, route guards)
- Forms & validation (Formik, Yup)
- Fetching data (fetch/axios), stateful vs stateless components

Module 7: Advanced React (Detailed)

- State management: Context API, Redux Toolkit (slices, thunks)
- Performance: memoization, code splitting, lazy loading, React Suspense
- Styling approaches: CSS Modules, Styled Components, Tailwind integration
- Error boundaries & graceful degradation
- Testing React components (Jest, React Testing Library)

Module 8: Full-Stack Integration (Detailed)

- Connecting React frontend to Express backend (CORS, proxy setups)
- Authentication & Authorization: JWT flows, refresh tokens, secure cookies, role-based routes
- File uploads: multer, streaming file upload best practices, presigned URLs for cloud storage
- Email & notification integration (nodemailer, third-party services)
- Payment gateway basics (Stripe/Razorpay integration)

Module 9: Testing & QA

- Unit testing (Jest), integration testing (supertest), E2E testing (Cypress)
- Test coverage, CI hooks, flaky test handling
- AI-assisted test generation (Copilot to scaffold tests)

Module 10: Modern MERN Additions

- TypeScript: benefits, types, generics, typing React props & Node types, migration tips
- GraphQL: schema design, resolvers, Apollo Server / Client basics
- Docker: Dockerfile, containerization, multi-stage builds, docker-compose
- CI/CD: GitHub Actions pipelines, automated testing, build & deploy flows
- Cloud Deployment: Vercel, Netlify, AWS (Elastic Beanstalk / ECS / Amplify), GCP, Render, Railway
- Basic System Design: scalability patterns, caching (Redis), rate limiting, horizontal scaling

Module 11: Project Development

- Designing & planning real-world app architecture
- Implementing CRUD, auth, role management
- Advanced features: search, filters, pagination, sorting
- Use cases: INoteBook, E-commerce platform, News App (as in PDF)

AI-Powered Development Tools (detailed) -

Purpose: How To Use AI Developer Tools To Accelerate Coding, Debugging, Testing, Documentation, And Product Features.

Tools & Subtopics (Hands-On)

1. GitHub Copilot / Copilot X

- Setup in VS Code, prompts & inline suggestions
- Use cases: scaffolding components, generating tests, refactoring, autocomplete large functions
- Hands-on: generate a React component + unit test with Copilot; evaluate & refine suggestions

2. Cursor AI

- Context-aware code edits, instant debug suggestions, multi-file understanding
- Hands-on: fix a failing API endpoint using Cursor prompts

3. Microsoft Copilot (IDE + GitHub Integrations)

- Enterprise features, PR summarization, code search, security suggestions

4. ChatGPT / Gemini For Developers

- Prompt engineering for code generation, explanations, bug reproduction steps, SQL/aggregation query help
- Use ChatGPT Code Interpreter / Advanced tools for generating sample data, test datasets

5. Tabnine / Codeium (Optional)

- Large-codebase autocomplete, private model integration

6. Replit / Warp AI (Developer Productivity Tools)

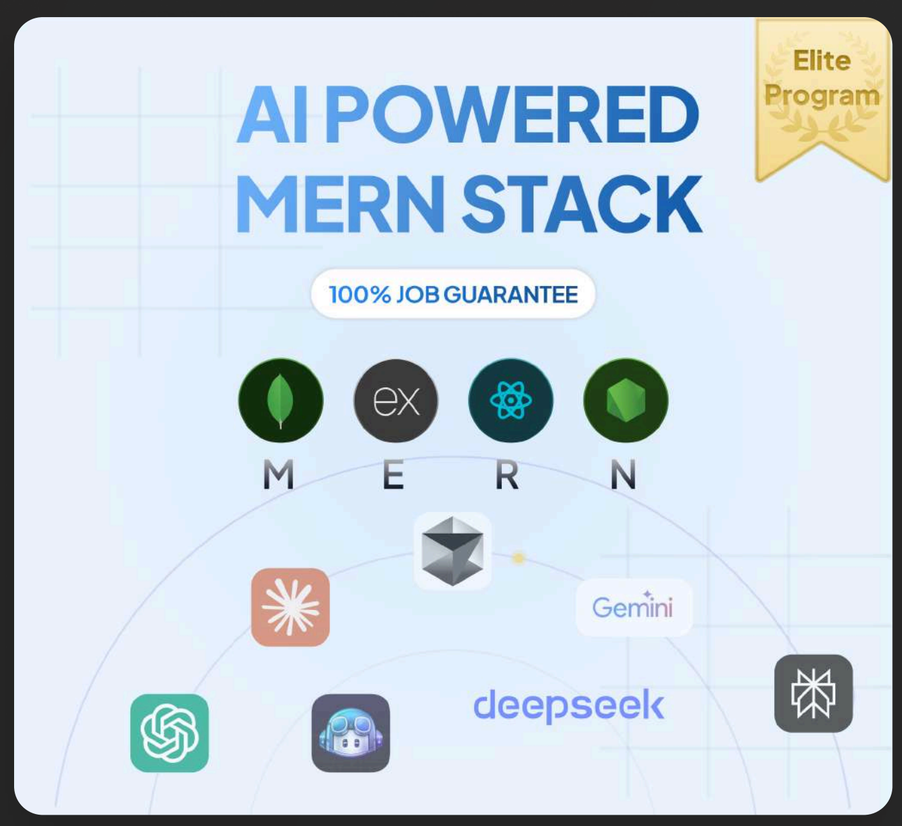
- Faster prototyping & terminal productivity

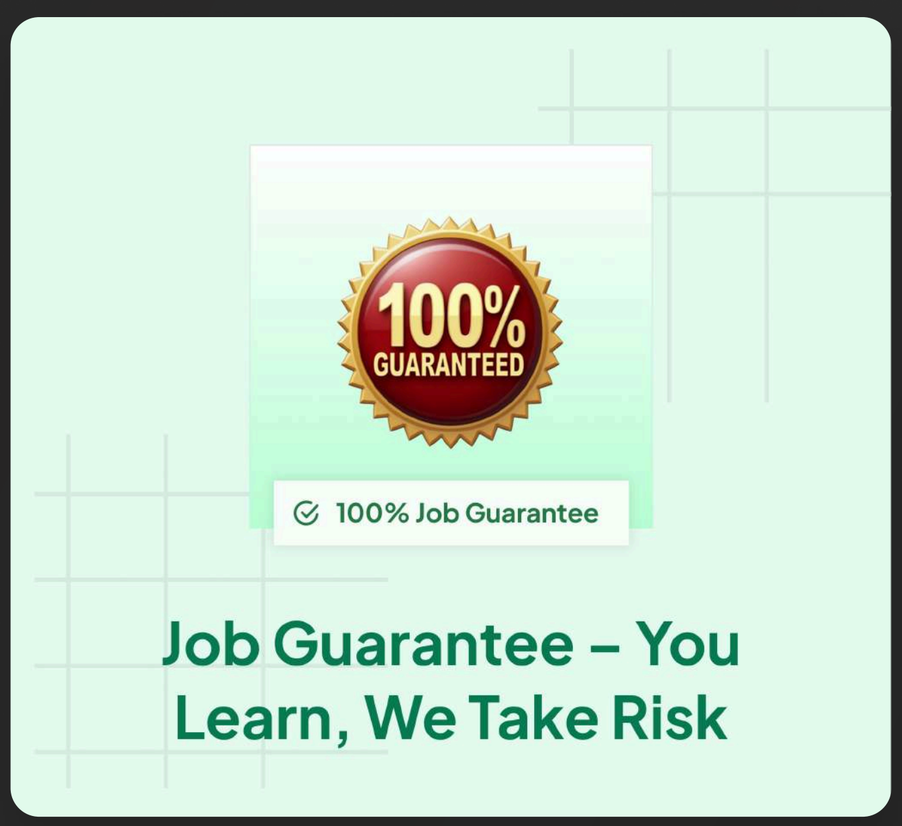
7. AI For Documentation & Tests

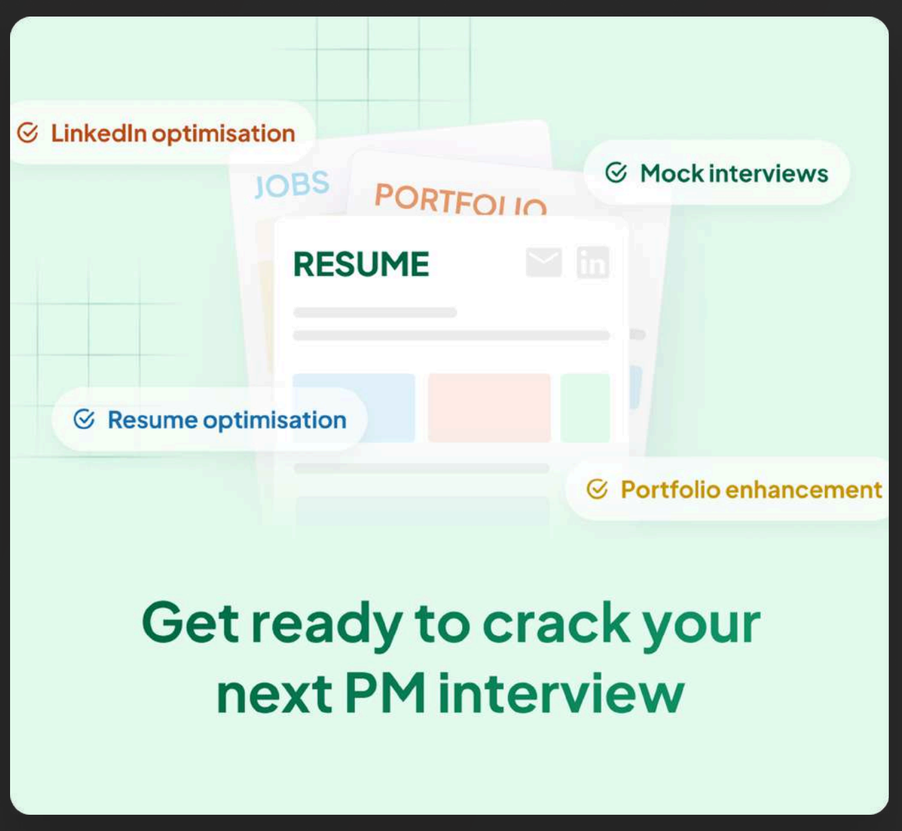
- Auto-generate README, API docs (OpenAPI), and test cases from code & comments

Practical Exercises (Per Tool):

- Prompt templates for component generation, API scaffolding, test generation
- Evaluate and edit AI output: teach students “verify, test, ship” workflow
- Security & license checks: spotting insecure AI suggestions











Total Program Fee

₹39,900/- ~~₹49,900/-~~

₹3,999/-	₹35,901/-
Registration fee	Payable After Registration
***Adjusted in Program fee*	